

Minoan names formed on plant words: KU-PA and cyperus, KU-MI-NA-QE and KI-KI-RA-JA in the Linear A archives

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Abstract

Three sign-group families of Linear A behave, in the administrative documents of Haghia Triada and Khania, as parties (persons, places or estates) and share their stem with a plant name of Pre-Greek or loan origin attested in Mycenaean Greek: KU-PA and its derivatives (Greek κύπαρος, Mycenaean ku-pa-ro, cyperus), KU-MI-NA-QE (ku-mi-no, cumin) and KI-KI-RA-JA, formed with the suffix -JA on KI-KI-NA, which Neumann identified in 1960 with the sycamore fig. The case of KU-PA is examined in detail: the sign-group recurs within single documents with quantities, its derivatives stand beside the logogram *303 on two roundels and on KH 5, the tablet headed by KA-KU-PA has the format of the Mycenae Ge spice records, and two of its affixes recur in the corpus above permutation level. The administrative context is reconstructed first: the documents of an aromatics administration, in which qualified oils (OLE+TA, OLE+RI, OLE+NE) and four further logograms cluster with *303, and the two sealing groups of the Haghia Triada roundels, in which a sign-group is paired with a commodity logogram. That context favours Younger's reading of the family as a party associated with cyperus deliveries rather than as the commodity itself; the reconciliation proposed is that the family is a name formed on the plant's word, a pattern familiar from Greek (Kyparissos, Marathon, Selinous) and here documented in the Minoan archives with two further cases. A screen of the whole vocabulary against seventy-nine phytonyms shows that two-syllable form matches alone carry no signal; the argument rests on the three cases and their three conditions each.

Keywords: Linear A, Minoan onomastics, phytonyms, cyperus, ku-pa-ro, Pre-Greek loanwords, roundels, distributional analysis.

1. Introduction

That Greek formed place-names and personal names on plant names is well known: Kyparissos and Kyparissia (κυπάρισσος, cypress), Marathon (μάραθρον, fennel), Selinous (σέλινον, celery), Rhodos (ρόδον, rose), Kydonia (the quince), and among personal names Hyakinthos, Narkissos, Daphne. Several of these phytonyms are themselves of Pre-Greek origin (Beekes 2010; Furnée 1972), and some are already attested in Mycenaean as place-names (ku-pa-ri-so at Pylos) or as commodities (ku-pa-ro, ku-mi-no, sa-sa-ma, ko-ri-ja-do-no in the Ga series of Knossos, the Ge series of Mycenae and the Un series of Pylos).

For Linear A the only accepted identification of a plant word is Neumann's (1960): KI-KI-NA on HT 88 as the sycamore fig (Greek κίκι, Egyptian kḳḳ), supported by the position of the sign-group after the fig logogram (NI · KI-KI-NA 7). Younger (2024a) lists in his lexicon dozens of sign-groups as "name in a list", "heading" or "place name?" without proposing an etymology for them, and in his commentary on HT 16 reads KA-KU-PA as the person responsible for the

contributions of the logogram *303. On the Mycenaean side, Jiménez Delgado (2024) has assembled the Pre-Greek names in -to of Linear B and compared them with Linear A sign-groups, relating ku-pa-nu-we-to to KU-PA₃-NU and KU-PA₃-NA-TU; his corpus shows that onomastic comparison between the two scripts is possible by etymology. Whether the names in the Minoan archives contain plant words has not been asked as such.

The interpretation of Linear A sign-groups rests on a small number of secure or widely accepted identifications obtained from the internal analysis of the documents: KU-RO ‘total’ and KI-RO ‘deficit’ (Duhoux 2011), a set of transaction terms and headings (SA-RA₂, KA-PA, A-DU; Schoep 2002; Montecchi 2010), and the toponyms and anthroponyms that recur in Knossian Linear B (Godart 1984). Identifications proposed on phonetic grounds alone, whatever the language of comparison, have not gained acceptance, for reasons set out by Duhoux (1989) and confirmed by recent computational work: with a syllabary of some ninety signs and sign-groups of two or three syllables, any sufficiently large lexicon yields matches (Nepal and Perono Cacciafoco 2024).

That observation can be quantified. For the 729 Linear A sign-groups with at least two consonants, read with the values of their Linear B homographs and reduced to a consonantal skeleton on a common inventory of thirteen classes, the proportion of skeletons attested in a comparison lexicon is 47% for Mycenaean Greek, 43% for Hittite, 62% for Ugaritic, 70% for Hebrew and 50% for Etruscan; against each lexicon with its consonants permuted, the ratios are 1.47, 1.46, 1.50, 1.68 and 1.57 respectively. No comparison language stands out, and the highest ratio corresponds to the largest lexicon. A method that starts from sound has no discriminating power on this corpus (Fig. 1).

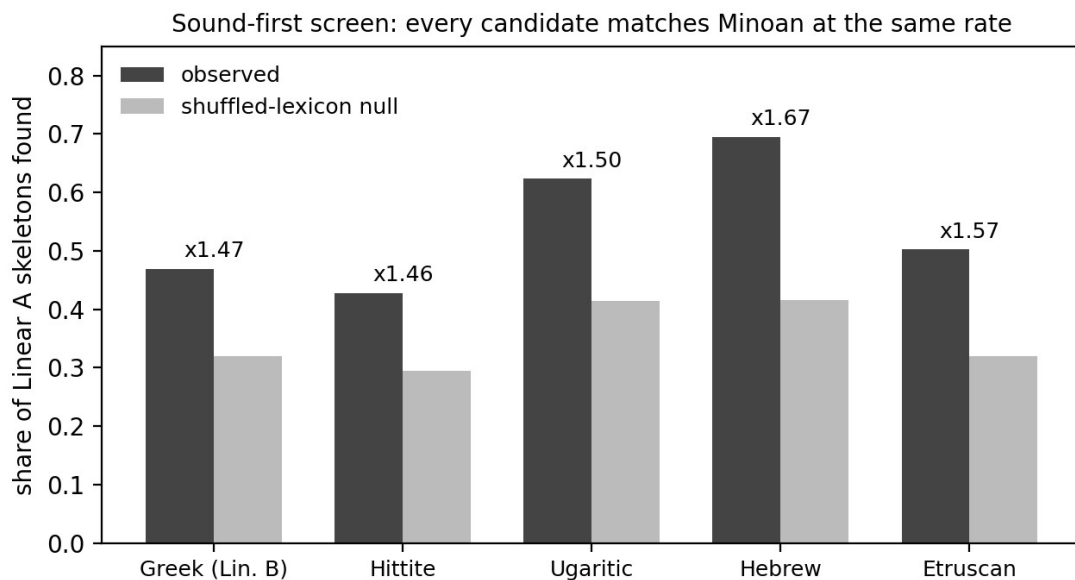


Fig. 1. Proportion of Linear A consonantal skeletons attested in five comparison lexicons, against permuted lexicons.

The procedure followed here is therefore the one that produced KU-RO: the function of a sign-group is established from its behaviour in the documents, and external comparison is admitted afterwards, as corroboration. Two additions are made. First, each distributional claim is tested against a permutation null (fixed-margin randomisation for co-occurrence, permutation of the affixed syllable for morphology); the same tests were applied to Etruscan, where the genitive before clan and the northern distribution of the marked sibilant are recovered (58% against a base rate of 26%; 0.6 against 0.1). Second, attention is paid to a behaviour that in an inventory separates a counted item from a name: recurrence within a single document with different quantities. The purpose of the calibration is the ordinary one: an instrument is tried first on material whose answer is known. Applied to Etruscan, where the genitive in -s or -l before clan and the northern distribution of the marked sibilant are established facts, the same tests recovered the genitive (58% of the words before clan, against a base rate of 26%), recovered the loss of internal vowels as a change over time that survives control by city, and dismissed an apparent change over time in the sibilant once city was controlled. Each null is a permutation of the data that preserves what must be preserved (the number of sign-groups in each document and the number of documents in which each sign-group occurs; or the inventory of final syllables), so that a claim of association is measured against what the same material would produce by chance. The procedure, its calibration and its negative results are described in a separate paper; the present note reports a single result.

2. Method

The starting point is the distributional analysis of sign-groups (position in the document, document type, accompanying logograms and quantities, stem / derivative pairs), tested against permutation nulls and calibrated on Etruscan, as described in section 1 and below. Formal comparison comes last and follows a fixed rule: agreement of the first two syllables, read with Linear B values, with a Pre-Greek or loan phytonym attested in Mycenaean or alphabetic Greek, allowing the U/E ~ O correspondence of final syllables shown by the Minoan names that survive at Knossos. For a match to count, three conditions are required: a shared stem; a party-like distribution (heading, entry in a personnel list, or roundel of the sealing group that pairs a sign-group with a commodity logogram, Weingarten 1987; Hallager 1996); and an attested derivative or a co-occurring logogram consistent with the plant.

As a control, the whole vocabulary (799 sign-groups) was screened against two lists of phytonyms, of sixty and of seventy-nine items, and the number of stem matches was compared with that obtained when syllables are permuted among sign-groups with their lengths preserved. With sixty phytonyms there are 91 matches against 53.1 expected (maximum 84 in 200 permutations); with seventy-nine, 97 against 73.7 (maximum 103). The excess of the first list is concentrated in the KU-PA family and disappears when the list is enlarged: the two-syllable rule has no discriminating power over the vocabulary as a whole, and the argument of this paper rests on the three individual cases and their three conditions, not on a global excess. The same screen applied to the Linear B corpus (5,597 sign-groups) produces 383 matches, most of them with

Greek personal names unrelated to plants (ke-re-te-u, ko-re-te), which confirms that a two-syllable match by itself is worthless.

3. The administrative context: aromatics and roundels

HT 16 deserves separate mention. Its entries are single signs used as commodity abbreviations followed by small fractions (WA $\frac{1}{3}$, *188+KU $\frac{3}{8}$, SA-PO ZE $\frac{1}{16}$), a format otherwise found in the Linear A tablets only on HT 20 (WA $\frac{1}{4}$, *188+KU $\frac{1}{2}$), written by the same hand as HT 16 in Godart's attribution (GORILA V, HT scribe 10), so that the format is in part a scribal habit (on the limits of palaeography as a criterion, Salgarella and Judson 2024); HT 30 and HT 32 are by HT scribe 1, and HT 23a, HT 35 and HT 60 have no attribution and, with logograms, on HT 23a, HT 32, HT 35 and HT 60. The format is that of the Mycenae Ge series, in which condiments are recorded by acrophonic abbreviation with the smallest measures of the system. A related observation concerns oil. The oil logograms of Linear A divide into two groups by their quantities and their company. OLE+U, plain OLE, OLE+KI, OLE+MI, OLE+DI and OLE+TU are recorded in whole units (medians between 3 and 7, up to 54) and share their documents with grain, olives, figs and *304, above chance in every case (hypergeometric tests on 1,719 documents): they are the oils of the estate's produce. OLE+TA, OLE+RI and OLE+NE are recorded in fractions and share their documents with *303 (OLE+TA in 4 of 7 documents, $p < 0.001$; OLE+RI and OLE+NE in 2 of 4 each, $p = 0.006$), with *308, MI+JA+RU and the ligatures QA₂+[*?]+PU and QA₂+[*?]+RE, and hardly ever with grain or olives: HT 23a, HT 30, HT 32, HT 35, HT 60, KH 61 and KH 85 belong to this group, and on HT 30 and HT 100 the allocation heading SA-RA₂ is followed by *303 with OLE+TA. The second group corresponds in function to the qualified oils of the Pylos Fr series (ku-pa-ro-we, wo-do-we, pa-ko-we) and to the condiment, perfume and dye plants that Sarpaki (2001) assembles from the Mycenaean texts and the archaeobotanical record, with the qualifier written as a ligatured sign rather than as a word, a device also used by Linear B in CYP+KU and PYC+KU, and one that the editors of the new ivory inscriptions from the Knossos cult centre describe in the same terms, syllabograms ligatured to vessel logograms as acrophonic qualifiers of type or content (Kanta, Nakassis, Palaima and Perna 2024); the values of TA, RI and NE are not known, and OLE+TA, the one most often recorded with *303, is the natural candidate for cyperus-scented oil.

The roundels of Haghia Triada fall into two sealing groups (Weingarten 1987; Hallager 1996; data from Younger's commentary, 2024b, and from SigLA). Those stamped by CMS II,6 no. 142 carry sign-groups without a commodity logogram (PA-SE-JA on two roundels, KA-RE-RO, TA-NI-KA, WA-O-MI, WA-DI-NI, ZU-DI-RA, A333-DI-NA-SU-KA), with two to four impressions each; those stamped by nos. 11, 21, 77 and 148 carry a sign-group followed by a commodity logogram: KA-KU-PA with *303 (no. 11, three impressions) and with *629 and *624 (no. 21, two impressions), QA-KU-RE with *303 ligatures (no. 21, five impressions), A-RA-TU-ME with OVIS f (no. 21, six impressions), KU-MI-NA-QE with CAP (no. 77, three impressions). The seal identifies the receiving official and not the party named, since one seal accompanies eight different sign-groups; in the second sealing group the sign-group is naturally the party and the logogram the commodity, since A-RA-TU-ME with ewes or KU-MI-NA-QE

with goats can hardly be commodity words. This is the frame within which the three cases below are read.

4. KU-PA and cyperus

4.1 The attestations of KU-PA

The readings below follow GORILA as tabulated in the LinearA Explorer (Hogan 2022); every attestation of the sign-group and of its derivatives has been checked against the SigLA dataset (Salgarella and Castellan 2020), which confirms AB 81 followed by AB 03 in all of them. The eleven documents in which KU is followed by AB 56 (pa₃) belong to a different sign-group, KU-PA₃-NU, a personal name with a Mycenaean counterpart (ku-pa-nu-we-to; Jiménez Delgado 2024 relates it also to KU-PA₃-NA-TU), and to KU-PA₃-RI-JA (HT 24a), on which see 4.3.

Document	Site	Support	Context
ZA 11a	Zakros	tablet	QE+GRA+PA 4 KU-PA · GRA+PA 1 PI-*310-A OLIV 2
ZA 11b	Zakros	tablet	I-OLIV 1 I-DE RA 2 KU-PA · GRA+PA 3
KH 29	Khania	tablet	RA · TA ₂ 1 KU- PA 2 *303 ½
HT 110a	Haghia Triada	tablet	SI-DU-*34-KU-MI *303+E 20 KU-PA 1 KU-RO 100

KU-PA is written on both faces of ZA 11 with the grain sign GRA+PA and the quantities 1 and 3, and at Khania with the quantity 2 in a document that also records *303. The quantity is in every case a small integer; no fraction follows the sign-group, although fractions follow the commodities recorded around it. Recurrence within a document is not in itself diagnostic of a commodity: the sign-groups that recur within documents are transaction terms (KU-RO, KI-RO), headings (MA-RU-ME on HT 24a, with 10 and 6) and, occasionally, names (DA-KU-SE-NE on HT 103, with 6 ½ and 1; KU-PA₃-NU twice on HT 122a). What the recurrence shows is that KU-PA is entered with quantities in more than one place in the same document; this is compatible with a counted commodity and with a party to whom grain is allotted twice. It must also be noted that the bare sign-group precedes a grain sign on ZA 11 and its own quantity on KH 29 and HT 110a; the association with *303 is carried by the derivatives, not by KU-PA itself, which may qualify a grain entry or record a particular presentation of the product (Figs. 3, 4).

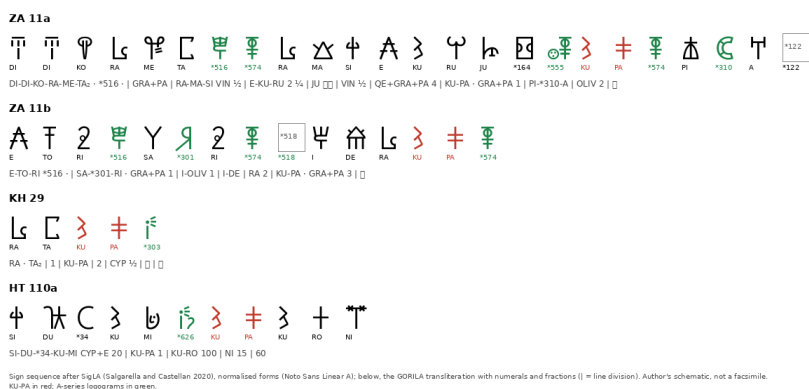


Fig. 3. The attestations of KU-PA on ZA 11a, ZA 11b, KH 29 and HT 110a: sign sequence after SigLA in normalised forms, with the GORILA transliteration. Author's schematic, not a facsimile.

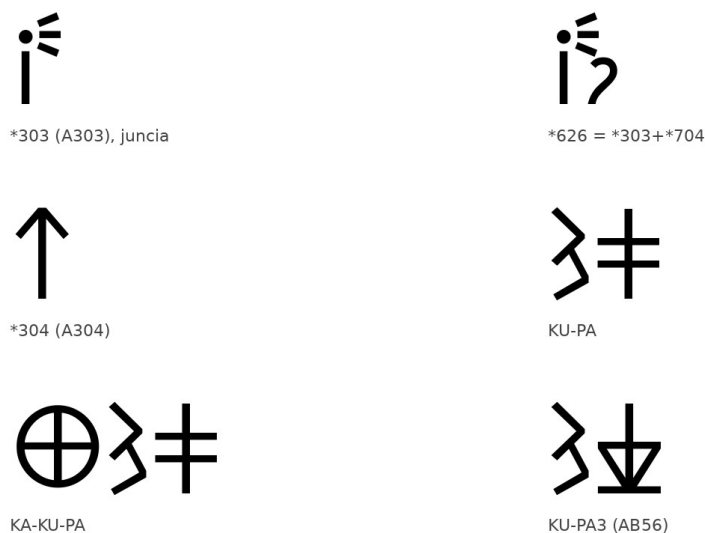


Fig. 4. The signs *303, *626, *304, and the sign-groups KU-PA, KA-KU-PA and KU-PA₃ in standard form.

4.2 The derivatives and the logogram *303

Sign-group	Document	Context
KA-KU-PA	HT Wc 3015 (roundel)	KA-KU-PA *303
KA-KU-PA	HT Wc 3016 (roundel)	KA-KU-PA *304+PA- *303+D
KA-KU-PA	HT 16 (tablet, heading)	KA-KU-PA · DI-NA-U ⅛ WA ⅓ *188+KU ⅔ SA-PO ZE ⅓ ₁₆

Sign-group	Document	Context
KU-PA-ZU	KH 5 (tablet)	KU-PA-ZU *303 3 $\approx \frac{1}{6}$ NI 2
*629 KU-PA	HT Wa 1020 (nodule)	*629 KU-PA $\frac{1}{4}$ JE-TA-NA
KU-PA-JA	HT 116a (tablet)	U-TA-RO · TE · KU-PA-JA GRA 16 PU-RA ₂ · GRA 40
KU-PA-RI	PE 1 (tablet)	U-KA-RE-A-SE-SI-NA KU-PA-RI VIR 50 GRA+PA 26 $\frac{1}{2}$

Two roundels from Haghia Triada bear KA-KU-PA followed only by *303 or by its ligature; KU-PA-ZU precedes *303 with a quantity at Khania; On the nodule HT Wa 1020 the sign-group follows the commodity logogram *629 (*304+PA) as two separate elements (SigLA reads *629, KU, PA): the nodule pairs a commodity sign with the sign-group rather than ligaturing them, and the transliteration *304+PA-KU-PA of the LinearA Explorer should not be read as a ligature. Of the twelve sign-groups that precede *303 anywhere in the corpus, two belong to this family. None of the family's attestations involves wine, oil, livestock or a vessel sign (Fig. 2).

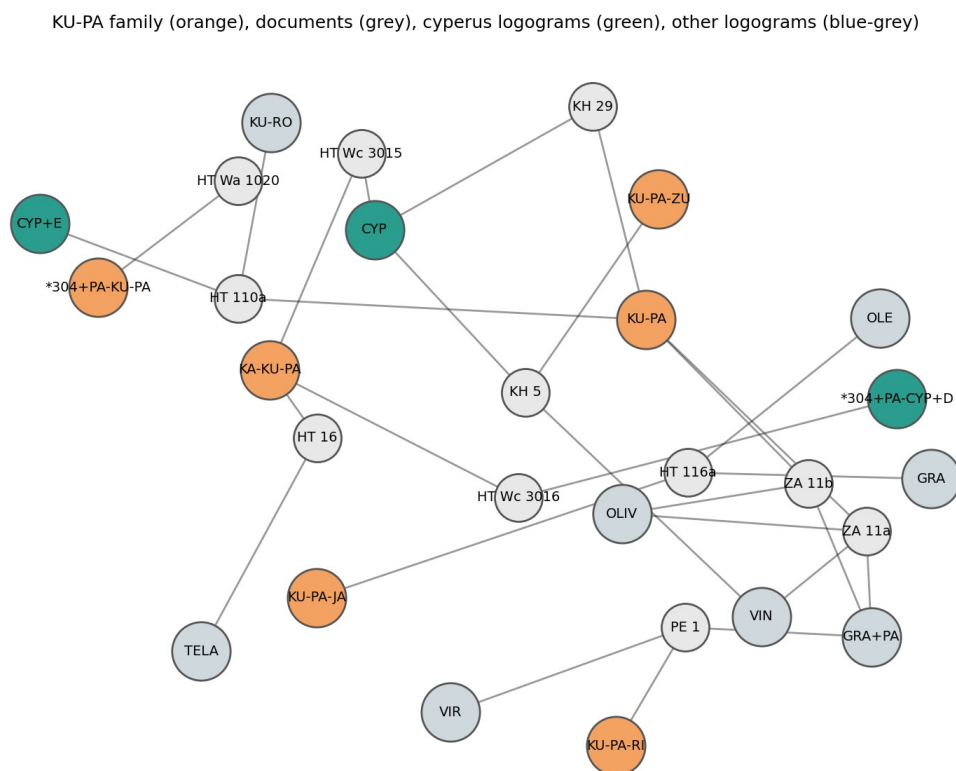


Fig. 2. Documents, logograms and quantities associated with KU-PA and its derivatives.

The reading of *303 (A 303) and of its ligatures (A 626, A 629) as cyperus is the usual one (Palmer 1995, apud Younger; on the iconographic design of the Aegean plant logograms, Weilhartner 2014) but is not secure: Del Frio, Nosch and Rougemont (2010) gloss A 626 as “cyperus / grain (?)”, and Schoep (2002, 93, apud Younger) records the disagreement between Hallager and others on which sign represents the plant. The argument below uses the conventional reading and, if accepted, supports it; the circularity is acknowledged and is broken only by the external comparison in section 3.

The documents in which the KU-PA family occurs are thus records of an administration of aromatics and scented oils, and the document headed by KA-KU-PA is, by its format, a list of condiments.

4.3 Younger’s reading and the sealing groups

Younger’s commentary on HT 16 already connects KA-KU-PA with *303, but in the opposite direction: KA-KU-PA, which “appears on two roundels, HT Wc 3015 and 3016, contributing *303”, is taken to be “the person responsible for the contributions of *303”. The Linear A Lexicon (Younger 2024a) glosses KU-PA as “name in a list?, transaction term?” (HT 110a.2, HT We 1020a, ZA 11a.5, b.3), KU-PA[on KH 29 as “name in a list”, KA-KU-PA as “(person’s name?)”, KU-PA-JA as “name in a list”, KU-PA-ZU as “third name in a three-name list”, KU-PA-RI as “word (placename?) in a list”, and KU-PA₃-RI-JA as “(from KU-PA-RI) heading, first word in a three-word list”. Neither cyperus nor ku-pa-ro is mentioned. Younger’s own hesitation over the base word, between a name and a transaction term, is resolved here towards the second alternative; and his derivation of KU-PA₃-RI-JA from KU-PA-RI implies an alternation of pa and pa₃ within the family, so that the sign-level distinction between KU-PA and KU-PA₃-NU holds for the signs while KU-PA₃-RI-JA, a heading, may belong with KU-PA-RI.

Whether the sign-group on a roundel names the commodity or the party to the sealed transaction is a question discussed by Hallager (1996), on whose analysis Younger’s reading depends. A roundel reading KA-KU-PA *303 is as naturally read as commodity as it is as contributor. The seal impressions do not decide the matter either: at Haghia Triada one seal, CMS II,6 no. 142, stamps the roundels bearing PA-SE-JA (HT Wc 3001, 3002), KA-RE-RO, ZU-NI-KA, TA-NI-KA, WA-O-MI, WA-DI-NI and A333-DI-NA-SU-KA alike (data from SigLA), so that the seal identifies the receiving official, as Hallager showed, and not the party or commodity named. The prefix KA- does not decide the matter (4.4); the behaviour of the base word does: it is counted, it stands beside a commodity sign on a nodule, its derivative KU-PA-ZU is followed by *303 and a quantity, and it never appears in a personnel list or with VIR. Two further observations bear on the choice. First, the sealing group described in section 3: KA-KU-PA shares seals nos. 11 and 21 with QA-KU-RE, A-RA-TU-ME and KU-MI-NA-QE, on roundels where the sign-group is the party and the logogram the commodity. Second, the entries of KH 5 are built as sign-group, logogram, quantity: WI-SA-SA-NE *626 2, KU-PA-ZU *303 3; two different sign-groups before two cyperus logograms in one tablet are more easily two parties than two words for the plant. These patterns favour Younger’s reading of the family’s function. They do not affect the

comparison in 4.5: a party named after the plant, a person, a place or an estate producing it, is what a name formed on the stem KU-PA would be, and Greek itself names places after plants (Kyparissos, Marathon, Selinous). The two readings are then compatible: KA-KU-PA a party named after the plant, KU-PA the plant's name. A coarse distributional profile (heading position, absence of VIR and of the quantity 1, presence on roundels) places the family, and KU-MI-NA-QE and KI-KI-RA-JA, closer to the known toponyms than to the known personal names; with one or two attestations per form this is an indication, not a proof, but it points to a place or estate rather than a person, as Younger already suggested for KU-PA₃-RI-JA ("perhaps a place near Petras"). The pattern is not isolated. KU-MI-NA-QE, which heads HT 89 and stands with CAP on the roundel HT Wc 3014 in the same sealing group as KA-KU-PA, shares its first two syllables with Mycenaean ku-mi-no, cumin; and KI-KI-RA-JA, which heads the personnel list of HT 85b, is formed with the suffix -JA on KI-KI-NA, the sign-group that Neumann (1960) identified with the sycamore fig and that is entered after the fig logogram on HT 88 (NI · KI-KI-NA 7); the two tablets are by different hands (HT scribes 7 and 9 in Godart's attribution), so that the relation between the two forms is not the habit of a single scribe. A screen of the whole vocabulary against twenty-six plant names of Pre-Greek or loan origin known from Mycenaean yields 37 stem matches against 18.6 expected when syllables are permuted (maximum 32 in 200 permutations); most of the excess is the KU-PA family itself, and two-syllable matches are individually weak, so the parallels are offered as a pattern to be tested, not as readings.

4.4 The affixes

Two affixes of the family recur in the corpus above the level expected under permutation of the affixed syllable. The prefix KA- forms five pairs of attested stem and prefixed form against 1.7 expected ($p = 0.013$): KA-KU-PA / KU-PA, KA-SA-RU / SA-RU, KA-NU-TI / NU-TI, KA-I-KA / I-KA, KA-RE-RO / RE-RO. The prefixed forms occur on roundels (HT Wc 3015, 3016, 3003; KN Wc 3) or in a tablet heading (HT 16); the bare forms are list entries with quantities (SA-RU 20 on HT 86, I-KA 5, NU-TI with VIR). The preference for roundels holds for the corpus as a whole: sign-groups beginning with ka- make up 17% of roundel sign-groups (5 of 29) against 4% of tablet sign-groups (29 of 697). The bases are names or place-names as often as commodities (KA-SA-RU is the form that survives in Knossian Linear B as ka-sa-ro), so KA-marks the document type rather than the class of referent; Younger (2010) had already noted prefixation and suffixation as the characteristic feature of roundel sign-groups. The suffix -JA forms seven pairs against 3.1 expected ($p = 0.020$): KU-PA-JA / KU-PA, PA-SE-JA / PA-SE, A-MA-JA / A-MA, A-RI-JA / A-RI, A-SE-JA / A-SE, JA-SA-JA / JA-SA, PU₂-RE-JA / PU₂-RE; in these pairs the bare form is a heading (PA-SE GRA+QE 20 on HT 18; A-MA on MA 1b and ZA 7b; A-SE on HT 81) and the suffixed form an entry with a quantity (KU-PA-JA GRA 16; PA-SE-JA 20 with VIR; A-MA-JA *303 6). The 49 sign-groups ending in -ja taken as a whole show no such distribution (41% in headings against a base rate of 34%), so the observation is confined to the pairs. -ZU, -RI and -RI-JA are too rare to test. The family may thus be described as a stem KU-PA, a roundel or heading form KA-KU-PA, a derived form KU-PA-JA used as an entry, and the longer forms KU-PA-RI and KU-PA₃-RI-JA.

4.5 Comparison: κύπαιρος, ku-pa-ro

Mycenaean cyperus has its own logograms, *124 PYC and *125 CYP, and the sign-group ku-pa-ro accompanies them in the records of aromatics. In the corpus used here: KN Ga(1) 517 tu-wi-no / ku-pa-ro AROM+PYC 1; KN Ga(1) 519 ku-pa-ro *171 2; KN Ga(3) 465 ni-so-de / ku-pa-ro T 1; PY Un 267 ku-pa-ro₂ AROM 6 (the perfumer's allocation); PY Un 249 ku-pa-ro₂ AROM 2 T 5; PY Fr 1203 ku-pa-ro-we wo-do-we OLE (see DMic s.v. ku-pa-ro; Aurora 2015). KN Ga 517, in which the word is followed by the cyperus logogram and a quantity, is the counterpart of HT Wc 3015 and 3016. The Linear B logogram itself carries the syllable KU as an acrophonic element in CYP+KU and PYC+KU; Linear A has, on the nodule HT Wa 1020, the commodity sign *629 followed by KU-PA (*304 being a grain sign, not *303). If *303 is the antecedent of *125, as the conventional reading assumes, the pairing of word and logogram is inherited rather than analogical.

Greek κύπειρον, κύπειρος, Ionic κύπερος, and κύπαιρος in Alcman denote the galingale, *Cyperus longus* and *C. rotundus*; Hesychius glosses κύπερα as ropes plaited from it. Frisk (GEW s.v.) regards the word as a foreign one of unknown origin whose fluctuating form reflects borrowing, and refers to Mycenaean ku-pa-ro; Beekes (2010 s.v.) classes it as Pre-Greek. The correspondence with the Linear A family is as close as the syllabary allows: KU-PA is the stem, KU-PA-RI (PE 1) supplies the -r- of κυπαρ-/κυπερ- and its passage to ku-pa-ro follows the same rule as the Minoan names that survive at Knossos (di-de-ru → di-de-ro, ka-sa-ru → ka-sa-ro: in 46 pairs with a counterpart identical but for the final vowel, Greek gives -o in 63%, and Minoan final -a never remains -a), and KU-PA₃-RI-JA, if Younger's derivation is accepted, supplies the form with yod reflected in ku-pa-ro₂ (kuparyos). The vowel fluctuation of the Greek forms is what would be expected from a source with an unstable second vowel. The identification of a Minoan plant name through a Greek gloss has precedents in Neumann (1960, 1962); the Minoan loan stratum in Mycenaean Greek was surveyed by Renfrew (1998); and the order of argument, function first and etymology afterwards, is that of Piquero (2015) for pa-ra-ku-ja, where a Minoan channel is also assumed. What is added here is not the etymology, which the dictionaries already give, but the Minoan sign-group itself, found in the Minoan documents beside the corresponding logogram, with a morphology that accounts for the Greek variants; whether the archives use the stem as the name of the plant or as the name of a party associated with it, the stem is the same.

An eastern source must also be considered: Semitic kpr (Ugaritic kpr, Hebrew kōfer), henna, an aromatic plant of the perfume trade, whence Greek κύπρος; a screen of the Linear A vocabulary against Akkadian, Ugaritic, Egyptian and Anatolian lexicons rendered in syllables (sections 2 and 5.4) yields no global excess over the null but recovers the three known sources (šamaššammū for sesame, kamūnu for cumin, Egyptian kꜥkꜥ for Neumann's sycamore fig) and pairs the KU-PA family with kpr. Whether KU-PA is the source of κύπειρον (cyperus, Pre-Greek) or a loan from kpr (henna) received by Greek as κύπρος does not alter the reading of the family as a name formed on an aromatic plant of the perfume administration; it alters the plant and the direction of borrowing, and only the identification of the logogram *303, which is

conventional, decides between the two. The question is left open. An alternative comparison must be considered. κύπελλον ‘cup’, with κύπη, κύμβη, Latin cūpa and Sanskrit kūpa, is a widely diffused word for containers, also regarded as Pre-Greek, and would suit KU-PA as a counted vessel or unit. Nothing in the context of KU-PA or of its derivatives is a vessel: the vessel names of Linear A are those of HT 31, written beside their pictograms, and none is built on ku-pa. The distribution favours the plant.

5. Two parallel cases

5.1 KU-MI-NA-QE and cumin

KU-MI-NA-QE occurs twice: heading HT 89 (a personnel tablet) and on the roundel HT Wc 3014, followed by the logogram CAP and stamped three times by CMS II,6 no. 77, in the same sealing group as the KA-KU-PA roundels (data from Younger 2024b and SigLA). A party-like distribution in both cases. The first two syllables agree with Mycenaean ku-mi-no, cumin, a Semitic loan (Akkadian kamūnu; Masson 1967; Rosół 2013) which Greek writes κύμινον. The element -NA-QE has no parallel in the KU-PA family; QE is frequent as a final syllable of roundel sign-groups (KU-MI-NA-QE, JA-*21F on HT 28a). The reading is a party called by the word for cumin.

5.2 KI-KI-RA-JA and the sycamore fig

KI-KI-NA is attested once, on HT 88, after the fig logogram and with the quantity 7: the plant or its fruit as a commodity, following Neumann’s identification (1960). KI-KI-RA-JA heads the personnel list of HT 85b (eleven entries with the quantity 1): the distribution of a party. The formal relation between the two is that of the suffix -JA, which in the corpus forms entries and derivatives from attested stems (seven pairs against 3.1 expected; 4.4), with the variant -RA-JA that Younger records in other sign-groups. The two tablets are by different hands (HT scribes 7 and 9 in Godart’s attribution). The case meets the three conditions and, unlike the other two, the identification of the plant is someone else’s and earlier.

5.3 Second tier

KA-NA heads HT 23a, the aromatics list, immediately before *303, and occurs on HT 123+124b and HT 135a; KA-NA-NI-TI is on a roundel from Khania with seven impressions (CMS V 233). The stem agrees with the first two syllables of Mycenaean ka-na-ko, safflower, the main product of the Ge lists; the third syllable is missing, and there is no KA-NA-KO in the corpus. The family KI-TA (KI-TA 2 on HT 30; KI-TA-I heading HT 39 with OLIV 31; KI-TA-NI-TE on a clay vessel) agrees with Mycenaean ki-ta-no, terebinth, whose resin and fruit were processed with oil; the co-occurrence with olives is suggestive, not probative. Both are left as candidates.

5.4 What the screen discards

The global excess of matches concentrates in the KU-PA family (fourteen matches) and in families that the two-syllable rule pairs with improbable phytonyms: KU-RO, the word for ‘total’, matches σκόροδον; the PI-TA family matches the pistachio, a Persian loan of classical date; the TU-ME family matches θύμον, of disputed etymology. Without the three conditions a

two-syllable match has no value, and this is stated. A specificity control by semantic field points the same way: with twenty-term lists of commodities attested in Mycenaean per field, only the plant field exceeds the maximum of its null (34 matches against a mean of 15.4 and a maximum of 25 in 200 permutations), while animals (15 against 8.1, maximum 18), vessels (16 against 9.3, maximum 17) and minerals (19 against 11.9, maximum 27) remain within chance; with the broad catalogue of seventy-nine phytonyms the excess disappears (section 2), so that the signal, where there is one, is specific to the plants that the Mycenaean administration records as commodities.

6. Discussion

The proposal has three supports: the consistent association of the family with the logogram *303 and with the documents of an aromatics administration, a morphology that can be described from the corpus, and an external form with the appropriate history. On the function of the family the evidence is divided: the recurrence of KU-PA with quantities and the format of HT 16 are compatible with a commodity, while the sealing groups of the roundels and the structure of KH 5 favour a party associated with cyperus deliveries, as Younger proposed. The reconciliation proposed here is that the family is a name formed on the plant's word. It is not a phonetic argument and has no bearing on the affiliation of the Minoan language. Its weakest points are the reading of *303, which is conventional rather than secure, and the fact that the bare sign-group is followed by a grain sign on ZA 11. The proposal can be tested: KU-PA and its derivatives should continue to appear with *303 and with dry commodities, in integer quantities, and not as a container of liquids; a new roundel with KA-KU-PA should carry *303 or a ligature of it; KU-PA-RI, if attested again, should accompany cyperus. Sesame (SA-SA-ME, HT 23a, recorded with the fraction $\frac{1}{16}$ in a document that also lists *303 and qualified oils; cf. Mycenaean sa-sa-ma in the Ge series) is the only other commodity term in the corpus that meets the same conditions; it is treated separately.

What is contributed is a pattern and not a decipherment: in the Minoan archives there are parties whose name contains the word for a plant, and that word is in all three cases one that Greek inherited as a loan (cyperus, cumin, sycamore fig). Three cases with three conditions each, two candidates and a control that shows where the signal ends. The pattern has a precedent in Greek itself and, in the case of KI-KI-RA-JA, the identification of the plant is Neumann's, so that the case does not depend on any new comparison.

Consequences for the reading of the archives. If the parties of the roundels and lists bear plant names, a portion of the sign-groups that Younger labels "name in a list" can be analysed as stem plus derivation, and the stem can be compared with the lexicon of Pre-Greek loans in the order followed here: function first, form afterwards. The suffix -JA and the prefix KA-, whose distributions were measured in 4.4, are the instruments of that decomposition.

What could refute the pattern: KU-MI-NA-QE or KI-KI-RA-JA appearing in a new document as a counted commodity with its own logogram, or KI-KI-NA appearing as a party; or the same stems appearing with incompatible distributions in the supplement to GORILA (Del Freo and

Zurbach 2024) or in the ivory inscriptions from the Knossos cult centre (Kanta, Nakassis, Palaima and Perna 2024).

The historical frame gives the proposal its context and its limits. The tablets and roundels of Haghia Triada, Khania and Zakros are the archive of the final year before the LM IB destructions (c. 1450), whatever their cause (Hood 1985; Driessen and Macdonald 1997; Wiener 2015): a moment, not a sequence, so that the twenty-one hands of Haghia Triada are colleagues rather than generations, and the commodities recorded may carry a seasonal bias. Knossos survived those destructions and there Linear B appears in Greek (Galanakis 2022); that the names of Haghia Triada reappear in Knossian Linear B and not at Pylos or Thebes (20 against 7 and 0), and that ku-pa-ro glosses on KN Ga 517 the logogram that KA-KU-PA accompanies on the roundels, is what is expected of an administration inherited in the same place with its personnel and vocabulary. The perfume industry documented at Pylos is, by contrast, a later Mycenaean development and is not projected onto the Minoan use of cyperus.

7. Ways forward

Three ways forward, in order of cost. First, the full catalogue of Pre-Greek phytonyms (Beekes 2010; Furnée 1972) instead of the lists used here, with the same rule and the same null; the excess should grow only in families with a party-like distribution. Second, the supplement to GORILA of 2024 and the Knossos inscriptions, which are not in the corpus used, for the five stems of this paper. Third, Mycenaean prosopography, for which Jiménez Delgado (2024) already provides the inventory of Pre-Greek names in -to: the personal names and ethnics of Knossos and Pylos formed on plants, established by etymology and not by screening (Landenius Enegren 2008; García Ramón 2011), such as ku-pa-ri-si-jo, the man from Kyparissos, on PY An 657, or the month name wo-de-wi-jo, that of the roses, on KN Fp 1, would give the term of comparison for whether the Minoan pattern continues in the Mycenaean administration of Crete, as the names of the group that survives at Knossos continue. Formal screening of Linear B is of no use for this, as shown in section 2; Greek etymology, which is available for Linear B, is.

Data

Transliterations: GORILA (Godart and Olivier 1976-1985) as tabulated by G. Douros and published in machine-readable form by the LinearA Explorer (Hogan 2022), which includes Younger's commentaries (2024b) with the seal data of the roundels and Godart's scribal attributions (GORILA V). Sign-level verification: the SigLA dataset (Salgarella and Castellan 2020, CC BY-NC-SA 4.0), in the form redistributed by pyaegian (sigla-corpus-v1). The Explorer labels the Zakros tablets, including ZA 11, as stone vessels; the support has been corrected from SigLA. Statistical tests and scripts accompany the paper.

Declaration of assistance

Corpus processing, permutation tests and a first draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction. All readings, comparisons and claims have been checked by the author, who is responsible for them.

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